Elements of Information Theory **2025-2026**

Lecture 1: Information Measures

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The goal is to introduce the basic measures of information on which we rely throughout the course.

Probability measure and preliminaries

In this course, we consider a probability space $(\Omega, \mathcal{H}, \mathrm{P})$ where Ω is the **sample space**, $\mathcal{H}$ is the **σ-algebra**, and P is the probability measure.

Let $(E, \mathcal{E})$ be a measurable space. A random variable X is a mapping $\Omega \to E$ that is measurable relative to $\mathcal{H}$ and $\mathcal{E}$. In particular, if E is countable, then we call X a **discrete random variable**. For any $A \in \mathcal{E}$, we can define $\mu(A) := \mathrm{P}(X(\omega) \in A)$, which is a probability measure on $(E, \mathcal{E})$. For discrete random variables, we call $\mathrm{P}_X(x) := \mathrm{P}(X(\omega) = x)$, $x \in E$, the **probability mass function (pmf)**.

Let μ and ν be two measures on a measurable space $(E, \mathcal{E})$, then ν is said to be **absolutely continuous** with respect to μ, denoted by $\nu \ll \mu$, if, for every set $A \in \mathcal{E}$, $\mu(A) = 0 \Rightarrow \nu(A) = 0$. If $\nu \ll \mu$, then there exists a Radon-Nikodym derivative of ν with respect to μ, often denoted by $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$, such that

$$\int_A \nu(\mathrm{d}x) = \int_A \mu(\mathrm{d}x) \frac{\mathrm{d}\nu}{\mathrm{d}\mu}(x), \quad \forall A \in \mathcal{E}.$$

Note that the Radon-Nikodym derivative is positive and measurable, i.e., in $\mathcal{E}_+$. If $\nu \ll \mu \ll \lambda$, we have $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\lambda} = \frac{\mathrm{d}\nu}{\mathrm{d}\lambda}$. If $\mathrm{P} \ll \mathrm{Q}$ are two probability measures defined on the same space $(E, \mathcal{E})$, and f is a P-measurable function, then we have the change of measure $\mathbb{E}_{\mathrm{P}} f(\mathrm{X}) = \mathbb{E}_{\mathrm{Q}}\big(f(\mathrm{X})\frac{\mathrm{dP}}{\mathrm{dQ}}(\mathrm{X})\big)$.

Consider the case where E of the random variable X is the Euclidean space. If the probability measure μ is absolutely continuous with respect to the Lebesgue measure, then $p_X(x) := \frac{\mathrm{d}\mu}{\mathrm{d}\lambda}(x)$ is called the **probability density function (pdf)**. We call the random variable X a **continuous random variable**.

The mapping $(x, B) \mapsto K(x, B)$, $x \in E$ and $B \in \mathcal{F}$, is a **transition kernel** from $(E, \mathcal{E})$ to $(F, \mathcal{F})$. In particular, we consider **probability transition kernel** such that $K(x, \mathcal{F}) = 1$ for all $x \in E$. If μ is a probability measure in E, then $\pi f = \int_E \mu(\mathrm{d}x) \int_F K(x, \mathrm{d}y) f(x, y)$ defines the unique probability measure satisfying $\pi(A \times B) = \int_A \mu(\mathrm{d}x) K(x, B)$ for all $A \in \mathcal{E}, B \in \mathcal{F}$. Conversely, under some regularity conditions, for every probability measure on the product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$, there exist a proability measure μ on E and a transition probability kernel K from $(E, \mathcal{E})$ to $(F, \mathcal{F})$ such that $\int_{E \times F} \pi(\mathrm{d}x \times \mathrm{d}y) f(x, y) =$

$\int_E \mu(\mathrm{d}x) \int_F K(x, \mathrm{d}y) f(x, y)$, also known as "disintegration". Throughout the course, we assume that such regularity conditions are met and ignore all measurability issues whenever possible. In most cases, we use $\mathrm{P}_{Y|X}$ to denote the transition probability kernel such that $\mathrm{P}_{Y|X=x}(\mathrm{d}y) = K(x, \mathrm{d}y)$. We use $\mathrm{P}_X \mathrm{P}_{Y|X}$ to denote the measure π on the product space such that $\int_{E\times F} \pi(\mathrm{d}x \times \mathrm{d}y)\, f(x, y) = \int_E \mathrm{P}_X(\mathrm{d}x) \int_F \mathrm{P}_{Y|X=x}(\mathrm{d}y)\, f(x, y)$. Both notations $\mathrm{P}_{Y|X}(\mathrm{d}y|x)$ and $\mathrm{P}_{Y|X=x}(\mathrm{d}y)$ are equivalent and can be used interchangeably. In particular, in the discrete case, $\mathrm{P}_{Y|X}(y|x) = \mathrm{P}_{Y|X=x}(y)$.

We use $\mathrm{P}_{Y|X} \circ \mathrm{P}_X$ to refer to the probability measure generated by the measure P_X and the transition kernel $\mathrm{P}_{Y|X}$:

$$(\mathrm{P}_{Y|X} \circ \mathrm{P}_X)(A) = \int_E \mathrm{P}_X(\mathrm{d}x) \mathrm{P}_{Y|X=x}(A).$$

It is the **marginalization** of the joint measure $\mathrm{P}_{Y|X}\mathrm{P}_X$. It can be regarded as the mixture of different distributions $\mathrm{P}_{Y|X=x}$ according to the measure P_X. In the discrete case, the transition probability kernel is a matrix and the pmf's are column vectors, and $\circ$ be be done as matrix multiplications. In most cases, we use P and Q to denote probability measures and p and q as the corresponding density function, i.e., pmf in the discrete case (w.r.t. the counting measure) and pdf in the continuous case (w.r.t. the Lebesgue measure). Finally, we remove the subscript of the pmf/pdf whenever ambiguity is not likely.

We will use the terms *distribution* and *probability measure* interchangeably. Unless the context makes it obvious, the underlying probability distribution will always be specified. Given a joint distribution P_{XY}, one can derive the marginals P_X and P_Y, as well as the conditional distributions (transition kernels) $\mathrm{P}_{Y|X}$ and $\mathrm{P}_{X|Y}$, such that

$$\mathrm{P}_{XY} = \mathrm{P}_X \mathrm{P}_{Y|X} = \mathrm{P}_Y \mathrm{P}_{X|Y}.$$

In this case, the notation $\mathrm{P}_X, \mathrm{P}_Y, \mathrm{P}_{Y|X}, \mathrm{P}_{X|Y}$ should always be understood as quantities induced by the given joint law P_{XY}. In more general situations, when several distributions or transition kernels are involved, we must be explicit about which objects are given and how others are constructed. For example, suppose we are given a conditional distribution $\mathrm{P}_{Y|X}$ but not a joint law. Together with two different input distributions P_X and Q_X, we can form two distinct joint distributions, $\mathrm{P}_{XY} := \mathrm{P}_X \mathrm{P}_{Y|X}, \mathrm{Q}_{XY} := \mathrm{Q}_X \mathrm{P}_{Y|X}$. In this case, P_{XY} and Q_{XY} are simply notations with explicit definition from the original distributions and transition kernels.

A real function f is called **convex** in a set $\mathcal{X}$ if for all $\lambda \in [0, 1]$ and all $x_1, x_2 \in \mathcal{X}$, we have

$$f(\lambda x_1 + (1 - \lambda) x_2) \le \lambda f(x_1) + (1 - \lambda) f(x_2),$$

which is quite easy to visualize. A real function is call **concave** if $-f$ is convex, or, equivalently, the above inequality changes direction. For example, $x \mapsto \log(x)$ is concave in $(0, \infty)$, $x \mapsto x \log(x)$ is convex in $(0, \infty)$.

One of the most important inequalities that we use in information theory is the so-called **Jensen's inequality**: Let $\mathrm{X} \in \mathcal{X}$ and f is convex in $\mathcal{X}$, then $\mathbb{E} f(\mathrm{X}) \ge f(\mathbb{E}\mathrm{X})$. For concave functions, we have $\mathbb{E} f(\mathrm{X}) \le f(\mathbb{E}\mathrm{X})$.

1.1 Entropy

Now, we introduce the first information measure. The **entropy** $H(\mathrm{X})$ of a discrete random variable $\mathrm{X} \sim \mathrm{P}_X$ is defined as

$$\boxed{H(\mathrm{X}) \equiv H(\mathrm{P}_X) := \mathbb{E}_{\mathrm{P}_X} \log \frac{1}{p(\mathrm{X})} = \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{p(x)}.}$$

Sometimes, we use the notation $H(\mathrm{P}_X)$ to emphasize that entropy is a functional of the pmf. Intuitively, entropy measures the *uncertainty* (or amount of *information*) of a random variable.

Similarly, we define the **joint entropy** with the joint distribution $\mathrm{P}_{X_1 \cdots X_n}$ (or, in short, P_{X^n}).

$$H(\mathrm{X}_1, \ldots, \mathrm{X}_n) := \mathbb{E}_{\mathrm{P}_{X^n}} \log \frac{1}{p(\mathrm{X}_1, \ldots, \mathrm{X}_n)} = \sum_{x_1 \in \mathcal{X}_1} \cdots \sum_{x_n \in \mathcal{X}_n} p(x_1, \ldots, x_n) \log \frac{1}{p(x_1, \ldots, x_n)}.$$

We also define the **conditional entropy** as

$$\boxed{H(\mathrm{X} \,|\, \mathrm{Y}) = H(\mathrm{P}_{X|Y} \mid \mathrm{P}_Y) := \mathbb{E}_{y \sim \mathrm{P}_Y} H(\mathrm{P}_{X|Y=y}) = \mathbb{E}_{\mathrm{P}_Y \mathrm{P}_{X|Y}} \log \frac{1}{p(\mathrm{X} \,|\, \mathrm{Y})}.}$$

Here Y need not be discrete. If Y is also discrete, then

$$H(\mathrm{X} \,|\, \mathrm{Y}) := \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p(x, y) \log \frac{1}{p(x \,|\, y)}.$$

For discrete $\mathrm{X}^n \sim \mathrm{P}_{X^n}$, we have the following **chain rule**:

$$\boxed{H(\mathrm{X}^n) = \sum_{i=1}^{n} H(\mathrm{X}_i \mid \mathrm{X}^{i-1})}$$

This can be shown with the chain rule of probability $p(x^n) = \prod_i p(x_i \mid x^{i-1})$:

$$\begin{aligned}
H(\mathrm{X}^n) &= \mathbb{E}_{\mathrm{P}_{X^n}} \left[\log \frac{1}{p(\mathrm{X}_1, \ldots, \mathrm{X}_n)} \right] \\
&= \sum_{i=1}^{n} \mathbb{E}_{\mathrm{P}_{X^n}} \left[\log \frac{1}{p(\mathrm{X}_i \mid \mathrm{X}^{i-1})} \right] \\
&= \sum_{i=1}^{n} \mathbb{E}_{\mathrm{P}_{X^i}} \left[\log \frac{1}{p(\mathrm{X}_i \mid \mathrm{X}^{i-1})} \right] \\
&= \sum_{i=1}^{n} H(\mathrm{X}_i \mid \mathrm{X}^{i-1})
\end{aligned}$$

Similarly, the conditional version also holds.

$$\boxed{H(\mathrm{X}^n \mid \mathrm{Y}) = \sum_{i=1}^{n} H(\mathrm{X}_i \mid \mathrm{X}^{i-1}, \mathrm{Y})}$$

1.2 Cross-entropy

Let us consider two probability measures P and Q defined on the same measurable space. Define the **cross-entropy** of P with respect to Q as[1]

$$H(\mathrm{P}\|\mathrm{Q}) := \begin{cases} \mathbb{E}_{\mathrm{P}} \log \frac{1}{q(\mathrm{X})}, & \mathrm{P} \ll \mathrm{Q} \\ +\infty, & \mathrm{P} \not\ll \mathrm{Q} \end{cases}$$

And the **conditional cross-entropy**

$$H(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y} \mid \mathrm{P}_Y) = \mathbb{E}_{y\sim \mathrm{P}_Y}\left[H(\mathrm{P}_{X|Y=y}\|\mathrm{Q}_{X|Y=y})\right]$$

When $\mathrm{P}_{X|Y=y} \ll \mathrm{Q}_{X|Y=y}$, P_Y-a.s., we have

$$\boxed{H(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y} \mid \mathrm{P}_Y) = \mathbb{E}_{\mathrm{P}_Y \mathrm{P}_{X|Y}}\left[\log \frac{1}{q(X|Y)}\right]}$$

As for entropy, the chain rule also holds for cross-entropy:

$$\boxed{H(\mathrm{P}_{X^n}\|\mathrm{Q}_{X^n}) = \sum_{i=1}^n H(\mathrm{P}_{X_i|X^{i-1}}\|\mathrm{Q}_{X_i|X^{i-1}} \mid \mathrm{P}_{X^{i-1}})}$$

$$\begin{aligned} H(\mathrm{P}_{X^n}\|\mathrm{Q}_{X^n}) &= \mathbb{E}_{\mathrm{P}_{X^n}}\left[\log \frac{1}{q(\mathrm{X}_1,\ldots,\mathrm{X}_n)}\right] \\ &= \sum_{i=1}^n \mathbb{E}_{\mathrm{P}_{X^n}}\left[\log \frac{1}{q(\mathrm{X}_i \mid \mathrm{X}^{i-1})}\right] \\ &= \sum_{i=1}^n \mathbb{E}_{\mathrm{P}_{X^i}}\left[\log \frac{1}{q(\mathrm{X}_i \mid \mathrm{X}^{i-1})}\right] \\ &= \sum_{i=1}^n H(\mathrm{P}_{X_i|X^{i-1}}\|\mathrm{Q}_{X_i|X^{i-1}} \mid \mathrm{P}_{X^{i-1}}) \end{aligned}$$

And the conditional version also holds.

$$\boxed{H(\mathrm{P}_{X^n|Y}\|\mathrm{Q}_{X^n|Y} \mid \mathrm{P}_Y) = \sum_{i=1}^n H(\mathrm{P}_{X_i|X^{i-1}Y}\|\mathrm{Q}_{X_i|X^{i-1}Y} \mid \mathrm{P}_{X^{i-1}Y})}$$

1.3 Differential entropy, differential cross-entropy

There is an equivalent definition of the entropy for continuous random variables. We define the **differential entropy** of a continuous random variable $\mathrm{X} \sim \mathrm{P}_X$ with pdf p as

$$\boxed{h(\mathrm{X}) \equiv h(\mathrm{P}_X) := \mathbb{E}_{\mathrm{P}_X}\left[\log \frac{1}{p(\mathrm{X})}\right] = \int \mathrm{d}x p(x) \log \frac{1}{p(x)}.}$$

[1] Here, we use a non-standard notation $H(\mathrm{P}\|\mathrm{Q})$ instead of $H(\mathrm{P},\mathrm{Q})$ to avoid confusion with the joint entropy $H(\mathrm{X},\mathrm{Y})$.

Similarly, we define the conditional differential entropy

$$h(\mathrm{X}\,|\,\mathrm{Y}) \equiv h(\mathrm{P}_{X|Y}\,|\,\mathrm{P}_Y) := \mathbb{E}_{y\sim \mathrm{P}_Y}\left[h(\mathrm{P}_{X|Y=y})\right] = \mathbb{E}_{\mathrm{P}_Y \mathrm{P}_{X|Y}}\left[\log \frac{1}{p(\mathrm{X}\,|\,\mathrm{Y})}\right].$$

Finally, if X^n has a density, we also define the **joint differential entropy**

$$h(\mathrm{X}^n) := \mathbb{E}\left[\log \frac{1}{p(\mathrm{X}^n)}\right].$$

The differential cross-entropy $h(\mathrm{P}\|\mathrm{Q})$ and the conditional version $h(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y}\,|\,\mathrm{P}_Y)$ are defined as the cross-entropy but with the pdf's.

The chain rule holds as for entropy.

1.4 Divergence

Let us now introduce the divergence (aka. Kullback-Leibler divergence, KL divergence, or relative entropy). Consider two probability measures P and Q defined on the same measurable space. The **divergence** of P from Q, denoted by $D(\mathrm{P}\|\mathrm{Q})$, is defined as

$$\boxed{D(\mathrm{P}\|\mathrm{Q}) := \begin{cases} \mathbb{E}_{\mathrm{P}}\left[\log \frac{\mathrm{dP}}{\mathrm{dQ}}(\mathrm{X})\right] = \mathbb{E}_{\mathrm{Q}}\left[\frac{\mathrm{dP}}{\mathrm{dQ}}(\mathrm{X}) \log \frac{\mathrm{dP}}{\mathrm{dQ}}(\mathrm{X})\right], & \mathrm{P} \ll \mathrm{Q} \\ +\infty, & \mathrm{P} \not\ll \mathrm{Q}. \end{cases}}$$

If P and Q are discrete distributions defined on the same set $\mathcal{X}$, then the divergence becomes

$$D(\mathrm{P}\|\mathrm{Q}) := \begin{cases} \mathbb{E}_{\mathrm{P}}\left[\log \frac{p(\mathrm{X})}{q(\mathrm{X})}\right] = \sum_{\mathcal{X}} p(x) \log \frac{p(x)}{q(x)}, & \mathrm{P} \ll \mathrm{Q} \\ +\infty, & \mathrm{P} \not\ll \mathrm{Q} \end{cases}$$

where the Radon-Nikodym derivative is simply the ratio of two pmf's.

If P and Q are distributions of continuous random variables defined on the same set $\mathcal{X}$, then the divergence becomes

$$D(\mathrm{P}\|\mathrm{Q}) := \begin{cases} \mathbb{E}_{\mathrm{P}}\left[\log \frac{p(\mathrm{X})}{q(\mathrm{X})}\right] = \int_{\mathcal{X}} \mathrm{d}x p(x) \log \frac{p(x)}{q(x)}, & \mathrm{P} \ll \mathrm{Q} \\ +\infty, & \mathrm{P} \not\ll \mathrm{Q} \end{cases}$$

where the Radon-Nikodym derivative is the ratio of two pdf's.

As for cross-entropy, one can also define the **conditional divergence** for an arbitrary probability measure P_Y and arbitary probability transition kernels $\mathrm{P}_{X|Y}$ and $\mathrm{Q}_{X|Y}$,

$$D(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y}\,|\,\mathrm{P}_Y) = \mathbb{E}_{y\sim \mathrm{P}_Y}\left[D(\mathrm{P}_{X|Y=y}\|\mathrm{Q}_{X|Y=y})\right]$$

When $\mathrm{P}_{X|Y=y} \ll \mathrm{Q}_{X|Y=y}$, P_Y-a.s., we have

$$\boxed{D(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y}\,|\,\mathrm{P}_Y) = \mathbb{E}_{\mathrm{P}_Y \mathrm{P}_{X|Y}}\left[\log \frac{p(X\,|\,Y)}{q(X\,|\,Y)}\right]}$$

Exactly as for cross-entropy, the chain rule of divergence holds:

$$\boxed{D(\mathrm{P}_{X^n}\|\mathrm{Q}_{X^n}) = \sum_{i=1}^{n} D(\mathrm{P}_{X_i|X^{i-1}}\|\mathrm{Q}_{X_i|X^{i-1}} \mid \mathrm{P}_{X^{i-1}})}$$

and the conditional version

$$D(\mathrm{P}_{X^n|Y}\|\mathrm{Q}_{X^n|Y} \mid \mathrm{P}_Y) = \sum_{i=1}^{n} D(\mathrm{P}_{X_i|X^{i-1}Y}\|\mathrm{Q}_{X_i|X^{i-1}Y} \mid \mathrm{P}_{X^{i-1}Y})$$

For discrete distributions, the following relation between the entropy, cross-entropy, and divergence is straightforward

$$H(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y} \mid \mathrm{P}_Y) = H(\mathrm{P}_{X|Y} \mid \mathrm{P}_Y) + D(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y} \mid \mathrm{P}_Y).$$

And in particular,

$$\boxed{D(\mathrm{P}\|\mathrm{Q}) = H(\mathrm{P}\|\mathrm{Q}) - H(\mathrm{P}).}$$

The same holds for the continuous counterpart, by replacing entropy/cross-entropy by differential entropy and differential cross-entropy.

1.5 Mutual information

Let P_{XY} be the **joint distribution** of (X, Y) (e.g. probability measure of the product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$. Similarly, let P_X and P_Y be the **marginal distributions** of X and Y, respectively. Further, let $\mathrm{P}_{X|Y}$ and $\mathrm{P}_{Y|X}$ be the transition probability kernels such that $\mathrm{P}_{XY} = \mathrm{P}_X\mathrm{P}_{Y|X} = \mathrm{P}_Y\mathrm{P}_{X|Y}$.[2]

Then, **mutual information** measures the dependence between X and Y:

$$\boxed{I(\mathrm{X};\mathrm{Y}) := D(\mathrm{P}_{XY}\|\mathrm{P}_X\mathrm{P}_Y) = D(\mathrm{P}_{X|Y}\|\mathrm{P}_X \mid \mathrm{P}_Y) = D(\mathrm{P}_{Y|X}\|\mathrm{P}_Y \mid \mathrm{P}_X),}$$

where the last two equalities can be proved with the chain rule of divergence. It compares the joint distribution to the one where X and Y are independent, or the conditional distributions $\mathrm{P}_{Y|X}$ and $\mathrm{P}_{X|Y}$ to the marginals P_Y and P_X. Sometimes mutual information is also denoted by $I(\mathrm{P}_X, \mathrm{P}_{Y|X})$, $I(\mathrm{P}_Y, \mathrm{P}_{X|Y})$, and $I(\mathrm{P}_{XY})$

$$\boxed{I(\mathrm{X};\mathrm{Y}) \equiv I(\mathrm{P}_{XY}) \equiv I(\mathrm{P}_X, \mathrm{P}_{Y|X}) \equiv I(\mathrm{P}_Y, \mathrm{P}_{X|Y}).}$$

Indeed, mutual information is a functional of the joint distribution P_{XY}.

In particular, in the discrete case

$$I(\mathrm{X};\mathrm{Y}) = \mathbb{E}_{\mathrm{P}_{XY}} \log \frac{p_{XY}(\mathrm{X},\mathrm{Y})}{p_X(\mathrm{X})p_Y(\mathrm{Y})}.$$

[2] We assume that both kernels exists, which is guaranteed when both $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are *standard spaces*.

It follows that

$$I(\mathrm{X};\mathrm{Y}) = H(\mathrm{X}) + H(\mathrm{Y}) - H(\mathrm{X},\mathrm{Y}) \tag{1.1}$$
$$= H(\mathrm{X}) - H(\mathrm{X}\,|\,\mathrm{Y}) \tag{1.2}$$
$$= H(\mathrm{Y}) - H(\mathrm{Y}\,|\,\mathrm{X}) \tag{1.3}$$

The relationship between entropy and mutual information is best visualized with the Venn diagram below.

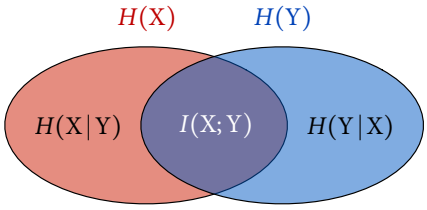


Figure 1.1: The Venn diagram.

In the continuous case, the definition is the same with the pdf

$$I(\mathrm{X};\mathrm{Y}) = \mathbb{E}_{\mathrm{P}_{XY}} \log \frac{p_{XY}(\mathrm{X},\mathrm{Y})}{p_X(\mathrm{X})p_Y(\mathrm{Y})}.$$

It follows that

$$I(\mathrm{X};\mathrm{Y}) = h(\mathrm{X}) + h(\mathrm{Y}) - h(\mathrm{X},\mathrm{Y}) \tag{1.4}$$
$$= h(\mathrm{X}) - h(\mathrm{X}\,|\,\mathrm{Y}) \tag{1.5}$$
$$= h(\mathrm{Y}) - h(\mathrm{Y}\,|\,\mathrm{X}), \tag{1.6}$$

similar to the discrete case.

If X is discrete and Y continuous, then

$$I(\mathrm{X};\mathrm{Y}) = D(\mathrm{P}_{XY}\|\mathrm{P}_X\mathrm{P}_Y) \tag{1.7}$$
$$= D(\mathrm{P}_{Y|X}\|\mathrm{P}_Y\,|\,\mathrm{P}_X) \tag{1.8}$$
$$= h(\mathrm{Y}) - h(\mathrm{Y}\,|\,\mathrm{X}). \tag{1.9}$$

We also have

$$I(\mathrm{X};\mathrm{Y}) = H(\mathrm{X}) - H(\mathrm{X}\,|\,\mathrm{Y}).$$

Note that in this case, although the conditional (differential) entropy exist, neither joint entropy nor joint differential entropy exists for (X,Y).

Let P_{XYZ} be some joint distribution of $(\mathrm{X},\mathrm{Y},\mathrm{Z})$. Then, we can define the **conditional mutual information** between X and Y given Z.

$$I(\mathrm{X};\mathrm{Y}\,|\,\mathrm{Z}) \equiv I(\mathrm{P}_{XY|Z}\,|\,\mathrm{P}_Z) := D(\mathrm{P}_{XY|Z}\|\mathrm{P}_{X|Z}\mathrm{P}_{Y|Z}\,|\,\mathrm{P}_Z) = D(\mathrm{P}_{X|YZ}\|\mathrm{P}_{X|Z}\,|\,\mathrm{P}_{YZ}) = D(\mathrm{P}_{Y|XZ}\|\mathrm{P}_{Y|Z}\,|\,\mathrm{P}_{XZ}),$$

where we replace the divergence in the definition of mutual information by the conditional divergence given Y.

The chain rule of mutual information is

$$\boxed{I(\mathrm{X};\mathrm{Y}^n) = \sum_{i=1}^n I(\mathrm{X};\mathrm{Y}_i \mid \mathrm{Y}^{i-1})}$$

Indeed, this can be proved from the (conditional) chain rule of divergence

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}^n) &= D(P_{Y^n|X}\|\mathrm{P}_{Y^n} \mid \mathrm{P}_X) \\ &= \sum_{i=1}^n D(P_{Y_i|XY^{i-1}}\|\mathrm{P}_{Y_i|Y^{i-1}} \mid \mathrm{P}_{XY^{i-1}}) \\ &= \sum_{i=1}^n I(\mathrm{X};\mathrm{Y}_i \mid \mathrm{Y}^{i-1}) \end{aligned}$$

The conditional version follows in the same way.

$$I(\mathrm{X};\mathrm{Y}^n \mid \mathrm{Z}) = \sum_{i=1}^n I(\mathrm{X};\mathrm{Y}_i \mid \mathrm{Y}^{i-1}\mathrm{Z})$$

1.6 Some properties of information measures

- General chain rule: Writing the chain rules in the same notational convention, we have

$$H(\mathrm{P}_{X^n}) = \sum_{i=1}^n H(\mathrm{P}_{X_i|X^{i-1}} \,|\, \mathrm{P}_{X^{i-1}}), \quad h(\mathrm{P}_{X^n}) = \sum_{i=1}^n h(\mathrm{P}_{X_i|X^{i-1}} \,|\, \mathrm{P}_{X^{i-1}})$$

$$H(\mathrm{P}_{X^n}\|\mathrm{Q}_{X^n}) = \sum_{i=1}^n H(\mathrm{P}_{X_i|X^{i-1}}\|\mathrm{Q}_{X_i|X^{i-1}} \,|\, \mathrm{P}_{X^{i-1}}), \quad h(\mathrm{P}_{X^n}\|\mathrm{Q}_{X^n}) = \sum_{i=1}^n h(\mathrm{P}_{X_i|X^{i-1}}\|\mathrm{Q}_{X_i|X^{i-1}} \,|\, \mathrm{P}_{X^{i-1}})$$

$$D(\mathrm{P}_{X^n}\|\mathrm{Q}_{X^n}) = \sum_{i=1}^n D(\mathrm{P}_{X_i|X^{i-1}}\|\mathrm{Q}_{X_i|X^{i-1}} \,|\, \mathrm{P}_{X^{i-1}})$$

$$I(\mathrm{P}_{X,Y^n}) = \sum_{i=1}^n I(\mathrm{P}_{XY_i|Y^{i-1}}|\mathrm{P}_{Y^{i-1}})$$

- Positivity

$$\begin{aligned} H(\mathrm{P}) \geq 0, \quad h(\mathrm{P}) \gtrless 0 \\ H(\mathrm{P}\|\mathrm{Q}) \geq 0, \quad h(\mathrm{P}\|\mathrm{Q}) \gtrless 0 \\ D(\mathrm{P}\|\mathrm{Q}) \geq 0 \Longrightarrow H(\mathrm{P}) \leq H(\mathrm{P}\|\mathrm{Q}),\ h(\mathrm{P}) \leq h(\mathrm{P}\|\mathrm{Q}) \\ I(\mathrm{P}_X, \mathrm{P}_{Y|X}) \geq 0 \end{aligned}$$

Proof. For entropy, since probability is upper bounded by 1, entropy is nonnegative. Entropy is 0 if and only if the random variable is deterministic. The positivity does not hold for differential entropy. The same arguments apply for cross-entropy and differential cross-entropy.

For divergence, if $\mathrm{P} \not\ll \mathrm{Q}$, then $D(\mathrm{P}\|\mathrm{Q}) = +\infty > 0$. We assume therefore $\mathrm{P} \ll \mathrm{Q}$. Then, let us write $D(\mathrm{P}\|\mathrm{Q}) = \mathbb{E}_{\mathrm{Q}}\left(f(\frac{\mathrm{dP}}{\mathrm{dQ}})\right)$ where $f(x) := x \log x$. Finally, since $f(x)$ is strictly convex (check), we have $D(\mathrm{P}\|\mathrm{Q}) \geq f(\mathbb{E}_{\mathrm{Q}} \frac{\mathrm{dP}}{\mathrm{dQ}}) = f(1) = 0$, where we applied Jensen's inequality on f. The equality holds if and only if $\frac{\mathrm{dP}}{\mathrm{dQ}}$ is constant (Q-almost everywhere), impling that $\mathrm{P} = \mathrm{Q}$.

The positivity of mutual information is from that of divergence. It is 0 if and only if $\mathrm{P}_{XY} = \mathrm{P}_X \mathrm{P}_Y$, i.e., X and Y are independent.

□

- Conditioning
  - Conditioning reduces (differential) entropy

$$H(\mathrm{X}) \geq H(\mathrm{X} \,|\, \mathrm{Y}), \quad h(\mathrm{X}) \geq h(\mathrm{X} \,|\, \mathrm{Y})$$

  - Conditioning increases divergence

$$D(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y} \mid \mathrm{P}_Y) \geq D(\tilde{\mathrm{P}}_X\|\tilde{\mathrm{Q}}_X)$$

    where $\tilde{\mathrm{P}}_X$ and $\tilde{\mathrm{Q}}_X$ are the marginals of $\mathrm{P}_{X|Y}\mathrm{P}_Y$ and $\mathrm{Q}_{X|Y}\mathrm{P}_Y$ respectively, i.e., $\tilde{\mathrm{P}}_X = \mathbb{E}_{y\sim\mathrm{P}_Y}[\mathrm{P}_{X|Y=y}]$ and $\tilde{\mathrm{Q}}_X = \mathbb{E}_{y\sim\mathrm{P}_Y}[\mathrm{Q}_{X|Y=y}]$.

Proof. Conditioning reduces entropy is from the positivity of mutual information, i.e., $H(\mathrm{X}) - H(\mathrm{X} \mid \mathrm{Y}) = I(\mathrm{X};\mathrm{Y}) \geq 0$. Similarly for differential entropy. For divergence, we have $D(\mathrm{P}_{X|Y}\|\mathrm{Q}_{X|Y} \mid \mathrm{P}_Y) = D(\mathrm{P}_{X|Y}\mathrm{P}_Y\|\mathrm{Q}_{X|Y}\mathrm{P}_Y) = D(\tilde{\mathrm{P}}_X\|\tilde{\mathrm{Q}}_X) + D(\tilde{\mathrm{P}}_{Y|X}\|\tilde{\mathrm{Q}}_{Y|X} \mid \tilde{\mathrm{P}}_X) \geq D(\tilde{\mathrm{P}}_X\|\tilde{\mathrm{Q}}_X)$, where we use the decompositions $\mathrm{P}_{X|Y}\mathrm{P}_Y = \tilde{\mathrm{P}}_X\tilde{\mathrm{P}}_{Y|X}$ and $\mathrm{Q}_{X|Y}\mathrm{Q}_Y = \tilde{\mathrm{Q}}_X\tilde{\mathrm{Q}}_{Y|X}$. □

- Convexity/Concavity
  - $\mathrm{P} \mapsto H(\mathrm{P})$, $\mathrm{P} \mapsto h(\mathrm{P})$ are both concave
  - $(\mathrm{P}, \mathrm{Q}) \mapsto D(\mathrm{P}\|\mathrm{Q})$ is convex
  - $\mathrm{P}_X \mapsto I(\mathrm{P}_X, \mathrm{P}_{Y|X})$ is concave
  - $\mathrm{P}_{Y|X} \mapsto I(\mathrm{P}_X, \mathrm{P}_{Y|X})$ is convex

Proof. Fix $\lambda \in [0,1]$. Let $\mathrm{S} \sim \mathrm{P}_S := \mathrm{Bern}(\lambda)$, i.e., $p_S(0) = \lambda$ and $p_S(1) = 1 - \lambda$.

Let $\mathrm{P}_{X|S=0} := \mathrm{P}_0$ and $\mathrm{P}_{X|S=1} := \mathrm{P}_1$. We have $\mathrm{P}_X = (1-\lambda)\mathrm{P}_0 + \lambda\mathrm{P}_1$. $\lambda H(\mathrm{P}_1) + (1-\lambda)H(\mathrm{P}_0) = H(\mathrm{X} \,|\, \mathrm{S}) \leq H(\mathrm{X}) = H(\mathrm{P}_X)$, proving the concavity of $\mathrm{P} \mapsto H(\mathrm{P})$ using conditioning reduces entropy.

For divergence, let $\mathrm{P}_{X|S=k} := \mathrm{P}_k$ and $\mathrm{Q}_{X|S=k} := \mathrm{Q}_k$ for $k = 0, 1$. Then, $\lambda D(\mathrm{P}_1\|\mathrm{Q}_1) + (1-\lambda)D(\mathrm{P}_0\|\mathrm{Q}_0) = (\mathrm{P}_{X|S}\|\mathrm{Q}_{X|S} \mid \mathrm{P}_S) \geq D(\mathrm{P}_X\|\mathrm{Q}_X)$, proving the convexity of divergence using conditioning increases divergence.

For the concavity of mutual information, for the given P_0, P_1, and $\mathrm{P}_{Y|X}$, let us set $\mathrm{P}_{X|S=0} := \mathrm{P}_0$ and $\mathrm{P}_{X|S=1} := \mathrm{P}_1$, and let $\mathrm{P}_{Y|XS} = \mathrm{P}_{Y|X}$, i.e., $\mathrm{P}_{Y|X,S=0} = \mathrm{P}_{Y|X,S=1} =$

$\mathrm{P}_{Y|X}$. Hence, the joint distribution is $\mathrm{P}_{SXY} = \mathrm{P}_S\mathrm{P}_{X|S}\mathrm{P}_{Y|X}$. The conditional mutual information

$$\begin{aligned} I(\mathrm{X};\mathrm{Y} \mid \mathrm{S}) &= D(\mathrm{P}_{XY|S}\|\mathrm{P}_{X|S}\mathrm{P}_{Y|S}|\mathrm{P}_S) \\ &= \lambda D(\mathrm{P}_{Y|X,S=0}\|\mathrm{P}_{Y|S=0} \mid \mathrm{P}_{X|S=0}) + (1-\lambda) D(\mathrm{P}_{Y|X,S=1}\|\mathrm{P}_{Y|S=1} \mid \mathrm{P}_{X|S=1}) \\ &= \lambda D(\mathrm{P}_{Y|X}\|\mathrm{P}_{Y|S=0} \mid \mathrm{P}_0) + (1-\lambda) D(\mathrm{P}_{Y|X}\|\mathrm{P}_{Y|S=1} \mid \mathrm{P}_1) \\ &= \lambda I(\mathrm{P}_0, \mathrm{P}_{Y|X}) + (1-\lambda) I(\mathrm{P}_1, \mathrm{P}_{Y|X}). \end{aligned}$$

On the other hand,

$$I(\mathrm{X};\mathrm{Y}) = I(\mathrm{P}_X, \mathrm{P}_{Y|X}) = I(\lambda\mathrm{P}_0 + (1-\lambda)\mathrm{P}_1, \mathrm{P}_{Y|X})$$

To finish the proof, we write

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}) &= I(\mathrm{X};\mathrm{Y}) + I(\mathrm{S};\mathrm{Y} \mid \mathrm{X}) \\ &= I(\mathrm{X},\mathrm{S};\mathrm{Y}) \\ &= I(\mathrm{S};\mathrm{Y}) + I(\mathrm{X};\mathrm{Y} \mid \mathrm{S}) \\ &\geq I(\mathrm{X};\mathrm{Y} \mid \mathrm{S}) \end{aligned}$$

where the first equality holds since $I(\mathrm{S};\mathrm{Y} \mid \mathrm{X}) = 0$ due to the Markov chain $\mathrm{S} \to \mathrm{X} \to \mathrm{Y}$. Indeed,

$$\begin{aligned} I(\mathrm{S};\mathrm{Y} \mid \mathrm{X}) &= D(\mathrm{P}_{Y|XS}\|\mathrm{P}_{Y|X} \mid \mathrm{P}_{XS}) \\ &= D(\mathrm{P}_{Y|X}\|\mathrm{P}_{Y|X} \mid \mathrm{P}_{XS}) \\ &= 0 \end{aligned}$$

Finally, for the convexity of mutual information, we need to prove that given P_X $I(\mathrm{P}_X, \lambda\mathrm{P}^1_{Y|X} + (1-\lambda)\mathrm{P}^0_{Y|X}) \leq \lambda I(\mathrm{P}_X, \mathrm{P}^1_{Y|X}) + (1-\lambda) I(\mathrm{P}_X, \mathrm{P}^0_{Y|X})$ for any kernels $\mathrm{P}^0_{Y|X}$ and $\mathrm{P}^1_{Y|X}$ and $\lambda \in [0,1]$. We can prove it in two different ways. First, we can apply the convexity of divergence. Indeed,

$$\begin{aligned} I(\mathrm{P}_X, \lambda\mathrm{P}^1_{Y|X} + (1-\lambda)\mathrm{P}^0_{Y|X}) &= D(\lambda\mathrm{P}_X\mathrm{P}^1_{Y|X} + (1-\lambda)\mathrm{P}_X\mathrm{P}^0_{Y|X}\|\lambda\mathrm{P}_X\mathrm{P}^1_Y + (1-\lambda)\mathrm{P}_X\mathrm{P}^0_Y) \\ &\leq \lambda D(\mathrm{P}_X\mathrm{P}^1_{Y|X}\|\mathrm{P}_X\mathrm{P}^1_Y) + (1-\lambda) D(\mathrm{P}_X\mathrm{P}^0_{Y|X}\|\mathrm{P}_X\mathrm{P}^0_Y) \\ &= \lambda I(\mathrm{P}_X, \mathrm{P}^1_{Y|X}) + (1-\lambda) I(\mathrm{P}_X, \mathrm{P}^0_{Y|X}) \end{aligned}$$

where P^0_Y and P^1_Y are the marginals of $\mathrm{P}_X\mathrm{P}^0_{Y|X}$ and $\mathrm{P}_X\mathrm{P}^1_{Y|X}$, respectively.

The second way is to introduce the same S as before, let $\mathrm{P}_{Y|X,S=0} = \mathrm{P}^0_{Y|X}$ and $\mathrm{P}_{Y|X,S=1} = \mathrm{P}^1_{Y|X}$, so that $\mathrm{P}_{SXY} = \mathrm{P}_S\mathrm{P}_X\mathrm{P}_{Y|XS}$. Unlike the previous cases, here X and S are independent. It can be verified that $\mathrm{P}_{Y|X} = \lambda\mathrm{P}^0_{Y|X} + (1-\lambda)\mathrm{P}^1_{Y|X}$. Therefore, we have $I(\mathrm{P}_X, \lambda\mathrm{P}^1_{Y|X} + (1-\lambda)\mathrm{P}^0_{Y|X}) = I(\mathrm{X};\mathrm{Y})$. We also have $\lambda I(\mathrm{P}_X, \mathrm{P}^1_{Y|X}) + (1-\lambda) I(\mathrm{P}_X, \mathrm{P}^0_{Y|X}) = I(\mathrm{X};\mathrm{Y} \mid \mathrm{S})$. Therefore, it is enough to prove that $I(\mathrm{X};\mathrm{Y} \mid \mathrm{S}) \geq I(\mathrm{X};\mathrm{Y})$. To that end, apply the independence so that $I(\mathrm{X};\mathrm{S}) = 0$, and thus

$$\begin{aligned} I(\mathrm{X};\mathrm{Y} \mid \mathrm{S}) &= I(\mathrm{X};\mathrm{Y} \mid \mathrm{S}) + I(\mathrm{X};\mathrm{S}) \\ &= I(\mathrm{X};\mathrm{Y},\mathrm{S}) \\ &= I(\mathrm{X};\mathrm{Y}) + I(\mathrm{S};\mathrm{Y} \mid \mathrm{X}) \\ &\geq I(\mathrm{X};\mathrm{Y}). \end{aligned}$$

□

- Data processing inequality (DPI)

$$D(\mathrm{P}_X\|\mathrm{Q}_X) \geq D(\mathrm{P}_{Y|X} \circ \mathrm{P}_X\|\mathrm{P}_{Y|X} \circ \mathrm{Q}_X)$$
$$I(\mathrm{P}_X, \mathrm{P}_{Y|X}) \geq I(\mathrm{P}_X, \mathrm{P}_{Z|Y} \circ \mathrm{P}_{Y|X})$$

where $\mathrm{P}_{Y|X} \circ \mathrm{P}_X$ denotes the marginal on Y from joint distribution $\mathrm{P}_{Y|X}\mathrm{P}_X$; $\mathrm{P}_{Z|Y} \circ \mathrm{P}_{Y|X}$ is the conditional kernel defined as $\{\mathrm{P}_{Z|Y} \circ \mathrm{P}_{Y|X=x} : x \in \mathcal{X}\}$. In other words, if $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$, we have

$$I(\mathrm{X};\mathrm{Y}) \geq I(\mathrm{X};\mathrm{Z}).$$

The conditional version of DPI also holds in the same way.

Proof. Let $\tilde{P}_Y := \mathrm{P}_{Y|X} \circ \mathrm{P}_X$ and $\tilde{Q}_Y := \mathrm{Q}_{Y|X} \circ \mathrm{P}_X$ be the marginals of Y from $\mathrm{P}_{Y|X}\mathrm{P}_X$ and $\mathrm{P}_{Y|X}\mathrm{Q}_X$, respectively. We have

$$\begin{aligned} D(\mathrm{P}_X\|\mathrm{Q}_X) &= D(\mathrm{P}_X\|\mathrm{Q}_X) + D(\mathrm{P}_{Y|X}\|\mathrm{P}_{Y|X} \mid \mathrm{P}_X) \\ &= D(\mathrm{P}_X\mathrm{P}_{Y|X}\|\mathrm{Q}_X\mathrm{P}_{Y|X}) \\ &= D(\tilde{\mathrm{P}}_Y\tilde{\mathrm{P}}_{X|Y}\|\tilde{\mathrm{Q}}_Y\tilde{\mathrm{Q}}_{X|Y}) \\ &= D(\tilde{\mathrm{P}}_Y\|\tilde{\mathrm{Q}}_Y) + D(\tilde{\mathrm{P}}_{X|Y}\|\tilde{\mathrm{Q}}_{X|Y} \mid \tilde{\mathrm{P}}_Y) \\ &\geq D(\tilde{\mathrm{P}}_Y\|\tilde{\mathrm{Q}}_Y) \end{aligned}$$

Obviously, the conditional version also holds similarly.

For the mutual information, since $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$, there exist $\mathrm{P}_{Z|Y}$ such that $\mathrm{P}_{Z|X} = \mathrm{P}_{Z|Y} \circ \mathrm{P}_{Y|X}$ and $\mathrm{P}_Z = \mathrm{P}_{Z|Y} \circ \mathrm{P}_Y$. Thus,

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}) &= D(\mathrm{P}_{Y|X}\|\mathrm{P}_Y \mid \mathrm{P}_X) \\ &\geq D(\mathrm{P}_{Z|Y} \circ \mathrm{P}_{Y|X}\|\mathrm{P}_{Z|Y} \circ \mathrm{P}_Y \mid \mathrm{P}_X) \\ &= D(\mathrm{P}_{Z|X}\|\mathrm{P}_Z \mid \mathrm{P}_X) \\ &= I(\mathrm{X};\mathrm{Z}). \end{aligned}$$

□

1.7 Maximum entropy

In the following, we show how to apply the property $H(\mathrm{P}) \leq H(\mathrm{P}\|\mathrm{Q})$ and $h(\mathrm{P}) \leq h(\mathrm{P}\|\mathrm{Q})$ to find out maximum entropy in different cases.

1.7.1 Finite alphabet

Let $|\mathcal{X}| = M < \infty$. Fix $\mathrm{Q} = \mathrm{Unif}(\mathcal{X})$. Then, for any distribution P_X over $\mathcal{X}$,

$$\begin{aligned} H(\mathrm{P}_X) &\leq H(\mathrm{P}_X\|\mathrm{Q}) \\ &= \mathbb{E}_{\mathrm{P}_X}[\log M] \\ &= \log M, \end{aligned}$$

where the equality holds when $\mathrm{P}_X = \mathrm{Q}$. Therefore, we show that uniform distribution maximizes entropy among all distributions with bounded aphabet size.

1.7.2 Continuous alphabet, finite second moment

Assume that P_X has a pdf and $\mathbb{E}\mathrm{X}^2 \le \sigma^2$. Fix $\mathrm{Q} \sim \mathcal{N}(0, \sigma^2)$, we have

$$\begin{aligned} h(\mathrm{P}_X) &\le h(\mathrm{P}_X \| \mathrm{Q}) \\ &= \mathbb{E}_{\mathrm{P}_X}\left[\log\sqrt{2\pi\sigma^2} + \frac{\mathrm{X}^2}{2\sigma^2}\log e\right] \\ &\le \frac{1}{2}\log(2\pi e\sigma^2), \end{aligned}$$

where the equalities hold when $\mathrm{P}_X = Q = \mathcal{N}(0, \sigma^2)$.

1.7.3 A general recipe

In general, if X has a density (e.g. pdf or pmf), then one can bound the (differential) entropy for a given expectation constraint $\mathbb{E}\left[c(\mathrm{X})\right] \le P$ for some positive function $x \mapsto c(x)$ such that the constraint can be satisfied with equality with some distribution.

Fix Q with

$$q(x) := \frac{e^{-\lambda c(x)}}{\int_{\mathcal{X}} e^{-\lambda c(x)}\mu(dx)} = \frac{1}{Z}e^{-\lambda c(x)},$$

where $\lambda \ge 0$ is such that $\mathbb{E}_{\mathrm{Q}}\left[c(\mathrm{X})\right] = P$.

Then, we have

$$\begin{aligned} h(\mathrm{P}_X) &\le h(\mathrm{P}_X \| \mathrm{Q}) \\ &= \mathbb{E}_{\mathrm{P}_X}\left[\log Z + \lambda\, c(\mathrm{X})\log e\right] \\ &\le \log Z + \lambda\, P \log e \end{aligned}$$

where the equalities holds when $\mathrm{P}_X = \mathrm{Q}$.

Exercises

1. One channel, two input distributions. Let $\mathcal{X} = \mathcal{Y} = \{0,1\}$ and $\varepsilon \in [0,1]$. The Z-channel $\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ is defined by $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=0} = \mathrm{Bern}(0)$ and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=1} = \mathrm{Bern}(1-\varepsilon)$: a 0 is always received correctly, and a 1 is received as 0 with probability ε. Let $\mathrm{P}_{\mathrm{X}} = \mathrm{Bern}(p)$ and $\mathrm{Q}_{\mathrm{X}} = \mathrm{Bern}(q)$.

   - Give the pmfs of the joint distributions $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ and $\mathrm{Q}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ as 2×2 tables, and the output distributions $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}}$ and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{Q}_{\mathrm{X}}$.
   - Compute the reverse channels of both joint distributions. Are they equal?
   - Show that $I(\mathrm{P}_{\mathrm{X}}, \mathrm{P}_{\mathrm{Y}|\mathrm{X}}) = H_2\big(p(1-\varepsilon)\big) - p\, H_2(\varepsilon)$. For $\varepsilon = \frac{1}{2}$, compare the values at $p = \frac{1}{2}$ and $p = \frac{2}{5}$.
   - For $p = \frac{1}{2}$, $q = \frac{1}{4}$ and $\varepsilon = \frac{1}{2}$, compute $D(\mathrm{P}_{\mathrm{X}} \| \mathrm{Q}_{\mathrm{X}})$ and $D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}} \| \mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{Q}_{\mathrm{X}})$, and check the data processing inequality.

2. Conditional measures are averages. Let $\mathrm{P}_{\mathrm{Y}} = \mathrm{Bern}(\frac{1}{2})$, $\mathrm{P}_{\mathrm{X}|\mathrm{Y}=0} = \mathrm{Bern}(\frac{1}{4})$ and $\mathrm{P}_{\mathrm{X}|\mathrm{Y}=1} = \mathrm{Bern}(\frac{3}{4})$, and let $\mathrm{Q}_{\mathrm{X}|\mathrm{Y}=0} = \mathrm{Q}_{\mathrm{X}|\mathrm{Y}=1} = \mathrm{Bern}(\frac{1}{2})$.

   - Compute $\mathrm{P}_{\mathrm{X}} := \mathrm{P}_{\mathrm{X}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}}$, $H(\mathrm{P}_{\mathrm{X}})$, $H(\mathrm{P}_{\mathrm{X}|\mathrm{Y}} \mid \mathrm{P}_{\mathrm{Y}})$ and $I(\mathrm{X};\mathrm{Y})$. Check that conditioning reduces entropy.
   - Compute $D(\mathrm{P}_{\mathrm{X}|\mathrm{Y}} \| \mathrm{Q}_{\mathrm{X}|\mathrm{Y}} \mid \mathrm{P}_{\mathrm{Y}})$ and $D(\mathrm{P}_{\mathrm{X}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}} \| \mathrm{Q}_{\mathrm{X}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}})$. Check that conditioning increases divergence.
   - Conditioning reduces entropy only on average. Give P_{Y} and $\mathrm{P}_{\mathrm{X}|\mathrm{Y}}$ such that $H(\mathrm{P}_{\mathrm{X}|\mathrm{Y}=y}) > H(\mathrm{P}_{\mathrm{X}})$ for some y.

3. Chain rules for sequences. Let $\mathrm{X}^n = [\mathrm{X}_1, \ldots, \mathrm{X}_n] \sim \mathrm{P}_{\mathrm{X}^n}$ with marginals $\mathrm{P}_{\mathrm{X}_1}, \ldots, \mathrm{P}_{\mathrm{X}_n}$ on a finite $\mathcal{X}$. For $\mathrm{P} \in \mathcal{P}(\mathcal{X})$, P^n denotes the product distribution, with pmf $\prod_{i=1}^n p(x_i)$.

   - Show that $H(\mathrm{P}^n) = nH(\mathrm{P})$ and $D(\mathrm{P}^n \| \mathrm{Q}^n) = nD(\mathrm{P} \| \mathrm{Q})$.
   - Show that $H(\mathrm{X}^n) \le \sum_{i=1}^n H(\mathrm{X}_i)$, with equality iff $\mathrm{X}_1, \ldots, \mathrm{X}_n$ are independent.
   - Let $\mathrm{Q}_1, \ldots, \mathrm{Q}_n \in \mathcal{P}(\mathcal{X})$, and let $\mathrm{Q}_{\mathrm{X}^n}$ have pmf $\prod_{i=1}^n q_i(x_i)$. Show that

$$\begin{aligned} D(\mathrm{P}_{\mathrm{X}^n} \| \mathrm{Q}_{\mathrm{X}^n}) &= D\big(\mathrm{P}_{\mathrm{X}^n} \| \mathrm{P}_{\mathrm{X}_1} \cdots \mathrm{P}_{\mathrm{X}_n}\big) + \sum_{i=1}^n D(\mathrm{P}_{\mathrm{X}_i} \| \mathrm{Q}_i) \\ &\ge \sum_{i=1}^n D(\mathrm{P}_{\mathrm{X}_i} \| \mathrm{Q}_i), \end{aligned}$$

     where $\mathrm{P}_{\mathrm{X}_1} \cdots \mathrm{P}_{\mathrm{X}_n}$ is the product of the marginals.

4. Erasure channel and convexity. In the binary erasure channel, $\mathcal{X} = \{0,1\}$, $\mathcal{Y} = \{0,1,\mathrm{e}\}$, and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$ gives $\mathrm{Y} = x$ with probability $1-\alpha$ and $\mathrm{Y} = \mathrm{e}$ (erasure) with probability α.

   - For $\mathrm{P}_{\mathrm{X}} = \mathrm{Bern}(p)$, give the pmf of $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}}$, and show that $I(\mathrm{P}_{\mathrm{X}}, \mathrm{P}_{\mathrm{Y}|\mathrm{X}}) = (1-\alpha)H_2(p)$. Which P_{X} maximizes it?
   - Show that the erasure channel is the mixture $(1-\alpha)\mathrm{P}^{(0)}_{\mathrm{Y}|\mathrm{X}} + \alpha \mathrm{P}^{(1)}_{\mathrm{Y}|\mathrm{X}}$ of the noiseless channel $\mathrm{P}^{(0)}_{\mathrm{Y}|\mathrm{X}}$ ($\mathrm{Y} = \mathrm{X}$) and the channel $\mathrm{P}^{(1)}_{\mathrm{Y}|\mathrm{X}}$ that always erases. Check the convexity of I in the channel here.

- For $\varepsilon \in [0, \frac{1}{2}]$, write the binary symmetric channel with crossover probability ε as a mixture of the binary symmetric channels with crossover probabilities 0 and $\frac{1}{2}$. Deduce from the convexity of I in the channel that $H_2(\varepsilon) \geq 2\varepsilon$.

5. Functions and data processing. Let X be discrete and g a function on $\mathcal{X}$.

   - Apply the chain rule to $H\big(\mathrm{X}, g(\mathrm{X})\big)$ in both orders to show that $H\big(g(\mathrm{X})\big) \leq H(\mathrm{X})$, with equality iff g is one-to-one on the support of $\mathrm{P_X}$. Compute $H(\mathrm{X})$ and $H(\mathrm{X}^2)$ for X uniform on $\{-2, -1, 0, 1, 2\}$.
   - For $\mathrm{P_X}, \mathrm{Q_X} \in \mathcal{P}(\mathcal{X})$, let $\mathrm{P}_{g(\mathrm{X})}$ and $\mathrm{Q}_{g(\mathrm{X})}$ be the distributions of $g(\mathrm{X})$ when $\mathrm{X} \sim \mathrm{P_X}$ and when $\mathrm{X} \sim \mathrm{Q_X}$. Show that $D(\mathrm{P}_{g(\mathrm{X})} \| \mathrm{Q}_{g(\mathrm{X})}) \leq D(\mathrm{P_X} \| \mathrm{Q_X})$. *Hint: write* $\mathrm{P}_{g(\mathrm{X})}$ *as the output distribution of a channel.*
   - Let $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$ be a Markov chain. Show that $I(\mathrm{X}; \mathrm{Z}) \leq I(\mathrm{Y}; \mathrm{Z})$, in addition to $I(\mathrm{X}; \mathrm{Z}) \leq I(\mathrm{X}; \mathrm{Y})$, and that $H(\mathrm{X} \,|\, \mathrm{Z}) \geq H(\mathrm{X} \,|\, \mathrm{Y})$.

6. Mixtures and the concavity of entropy. Let $\mathrm{P}_0, \mathrm{P}_1 \in \mathcal{P}(\mathcal{X})$ on a finite $\mathcal{X}$, $\lambda \in [0, 1]$ and $\mathrm{P}_\lambda := (1 - \lambda)\mathrm{P}_0 + \lambda \mathrm{P}_1$. As in the lecture, let $\mathrm{S} \sim \mathrm{Bern}(\lambda)$ and $\mathrm{P}_{\mathrm{X}|\mathrm{S}=s} = \mathrm{P}_s$ for $s \in \{0, 1\}$.

   - Show that $\mathrm{P}_{\mathrm{X}|\mathrm{S}} \circ \mathrm{P_S} = \mathrm{P}_\lambda$ and that $H(\mathrm{P}_\lambda) - (1 - \lambda)H(\mathrm{P}_0) - \lambda H(\mathrm{P}_1) = I(\mathrm{S}; \mathrm{X})$.
   - Show that $I(\mathrm{S}; \mathrm{X}) = (1 - \lambda)D(\mathrm{P}_0 \| \mathrm{P}_\lambda) + \lambda D(\mathrm{P}_1 \| \mathrm{P}_\lambda)$.
   - Deduce that $0 \leq H(\mathrm{P}_\lambda) - (1 - \lambda)H(\mathrm{P}_0) - \lambda H(\mathrm{P}_1) \leq H_2(\lambda)$. When does equality hold on each side?

7. Maximum entropy distributions. Use the Gibbs distribution of the lecture, with the constraint $\mathbb{E}_\mathrm{P}\big[c(\mathrm{X})\big] \leq a$ for a given $a > 0$.

   - On $\mathcal{X} = \{0, 1, 2, \ldots\}$, find the pmf that maximizes $H(\mathrm{P})$ under $\mathbb{E}_\mathrm{P}[\mathrm{X}] \leq a$, and show that the maximum is $(1 + a)\log(1 + a) - a \log a$.
   - On $\mathcal{X} = [0, \infty)$, find the pdf that maximizes $h(\mathrm{P})$ under $\mathbb{E}_\mathrm{P}[\mathrm{X}] \leq a$, and the maximum.
   - On $\mathcal{X} = \mathbb{R}$, find the pdf that maximizes $h(\mathrm{P})$ under $\mathbb{E}_\mathrm{P}\big[|\mathrm{X}|\big] \leq a$, and the maximum.
   - On $\mathcal{X} = \mathbb{R}$, show that $h(\mathrm{P})$ is unbounded under $\mathbb{E}_\mathrm{P}[\mathrm{X}] \leq a$. Which assumption of the lecture fails?

8. Cross-entropy as a loss. A label $\mathrm{Y} \in \mathcal{Y}$ is predicted from a feature $\mathrm{X} \in \mathcal{X}$, both finite, with $(\mathrm{X}, \mathrm{Y}) \sim \mathrm{P_X P_{Y|X}}$. A model $\mathrm{Q_{Y|X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$ is chosen to minimize the expected log-loss $L(\mathrm{Q_{Y|X}}) := \mathbb{E}_{\mathrm{P_X P_{Y|X}}}\big[\log \frac{1}{q(\mathrm{Y} \,|\, \mathrm{X})}\big]$.

   - Show that $L(\mathrm{Q_{Y|X}}) = H(\mathrm{P_{Y|X}} \| \mathrm{Q_{Y|X}} \mid \mathrm{P_X}) = H(\mathrm{P_{Y|X}} \mid \mathrm{P_X}) + D(\mathrm{P_{Y|X}} \| \mathrm{Q_{Y|X}} \mid \mathrm{P_X})$.
   - Deduce that $L(\mathrm{Q_{Y|X}}) \geq H(\mathrm{Y} \,|\, \mathrm{X})$, with equality iff $\mathrm{Q}_{\mathrm{Y}|\mathrm{X}=x} = \mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$ for every x with $p(x) > 0$.
   - A model that ignores the feature has $\mathrm{Q}_{\mathrm{Y}|\mathrm{X}=x} = \mathrm{Q_Y}$ for every x. Show that the best such model is $\mathrm{Q_Y} = \mathrm{P_{Y|X}} \circ \mathrm{P_X}$, and that its loss exceeds the smallest loss by $I(\mathrm{X}; \mathrm{Y})$.